Flow of Incompressible Fluids in Pipes and channels

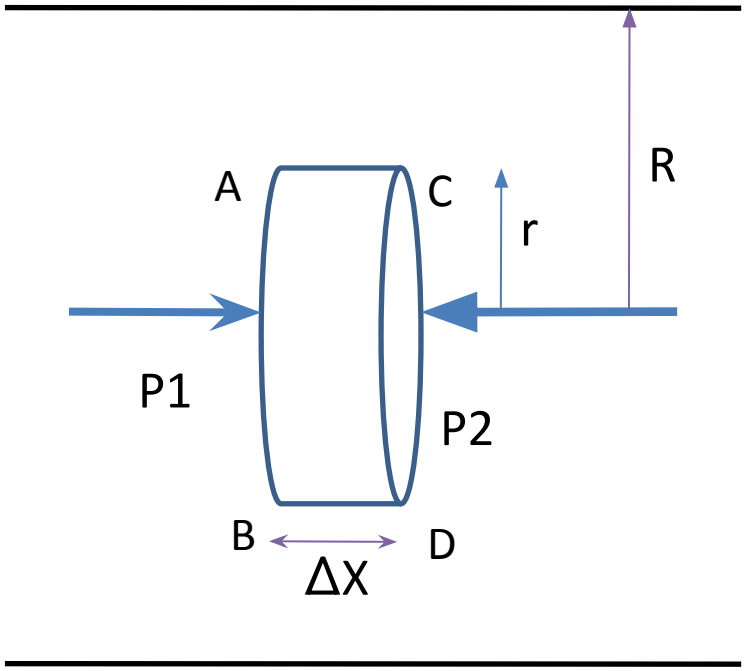
- ✔ Shear Stress and skin friction in Pipes
- ✔ Laminar Flow in Pipes and Channels
- ✔ Velocity profile and shear stress distribution in laminar flow
- ✔ Definition of friction factor
- ✔ Friction loss from sudden expansion of cross section
- ✔ Flow of liquids in thin layers over a flat plate
- ✔ Turbulent flow in pipes and channels

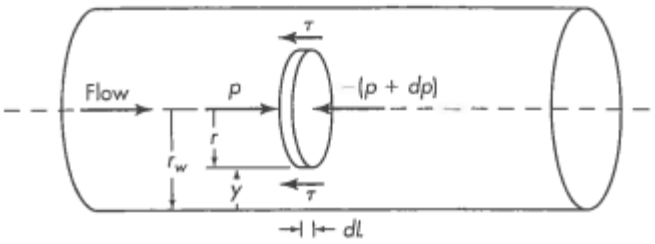


Flow of Incompressible Fluids in Pipes

- Fluid flowing in pipes has two primary flow patterns.
- It can be either laminar when all of the fluid particles flow in parallel lines at even velocities and it can be turbulent when the fluid particles have a random motion interposed on an average flow in the general direction of flow
- consider the steady laminar flow of an incompressible fluid with constant properties in the fully developed region of a straight circular pipe
- In fully developed laminar flow, each fluid particle moves at a constant axial velocity along a streamline and the velocity profile u(r) remains unchanged in the flow direction. There is no motion in the radial direction, and thus the velocity component in the direction normal to flow is everywhere zero. There is no acceleration since the flow is steady and fully developed.

Flow of Incompressible Fluids in Pipes





- Consider a horizontal pipe of radius R in which a steady flow of a viscous fluid at constant density is taking place.
- Visualize a disk shaped element of fluid, concentric with the axis of the tube of radius r and length Δx as shown in figure

- Let the fluid pressure on the upstream and downstream faces of the disk be (ie., faces AB and CD) P_1 and P_2 respectively.
- The volume element involves only pressure and viscous effects and thus the pressure and shear forces must balance each other.
- Since the fluid posses a viscosity, a shear force opposing the flow will exist on the rim of the element.

The forces acting on the fluid element are

- pressure force, $P_1 (\pi r^2)$ on the face AB
- pressure force, $P_2(\pi r^2)$ on the face CD
- shear force τ ($2 \pi r \Delta x$) on the surface of fluid element.

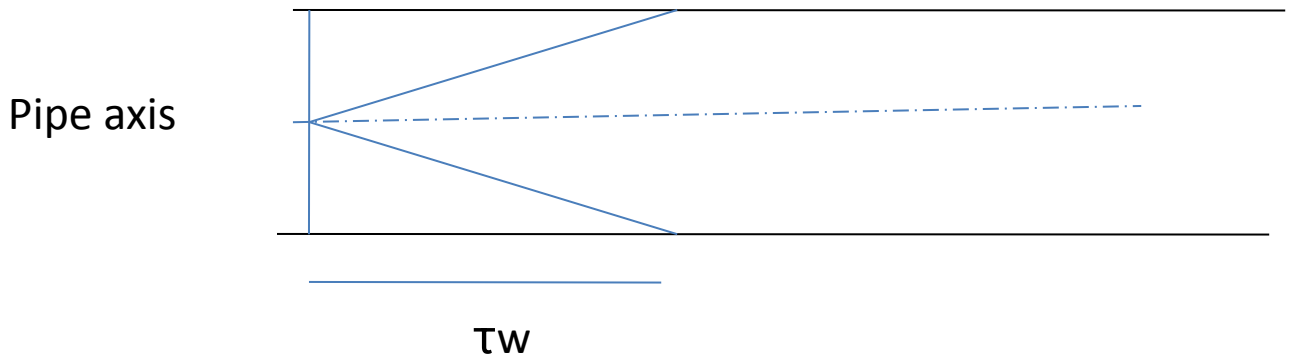
As there is no acceleration, the summation of all forces in the direction of flow must be zero

i.e., $P_1 \pi r^2 - P_2 \pi r^2 - \tau 2 \pi r \Delta x = 0$

$(P_1 - P_2) \pi r^2 - \tau 2 \pi r \Delta x = 0$

$$\tau = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r}{2} \rightarrow (1)$$

The shear stress, τ across a section varies with r since $\left(\frac{P_1 - P_2}{\Delta x} \right)$ is constant shear stress distribution across a section is linear as shown in the figure.



Expression to find Velocity distribution across a pipie

From force balance shear stress variation is

$$\tau = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r}{2} \rightarrow (1)$$

from Newton's law of viscosity

$$\tau = - \mu \frac{du}{dr} \rightarrow (2)$$

Equating (1) and (2)

$$-\mu \frac{du}{dr} = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r}{2}$$

$$-du = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r}{2\mu} dr$$

, integrating with respect to r gives

$$-u = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r^2}{4\mu} + c \rightarrow (3)$$

where c is the integration constant

its value is obtained from the boundary conditions

i.e., at r= R ; u=0

$$0 = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{R^2}{4\mu} + c$$

$$\Rightarrow c = - \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{R^2}{4\mu}$$

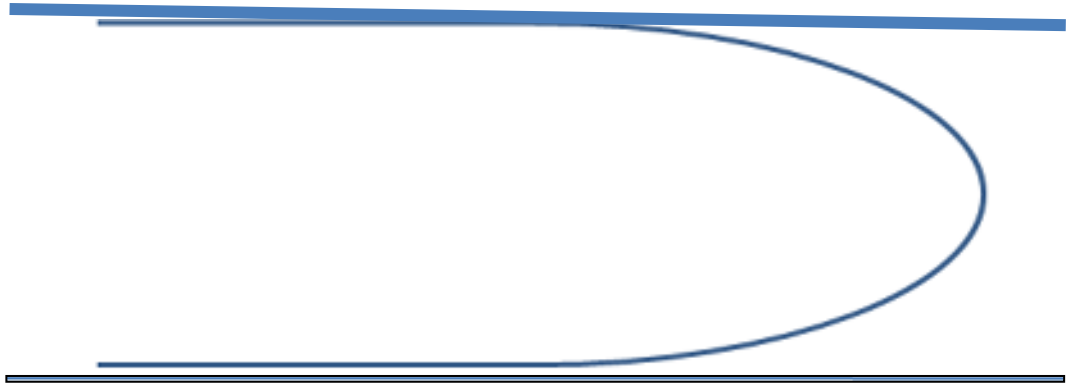
substituting the value of c in the equation (3)

$$- u = \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{r^2}{4\mu} - \left(\frac{P_1 - P_2}{\Delta x} \right) \frac{R^2}{4\mu}$$

$$u = \frac{1}{4\mu} \left(\frac{P_1 - P_2}{\Delta x} \right) (R^2 - r^2) \rightarrow (4)$$

In the above equation the values of μ, $\left(\frac{P_1 - P_2}{\Delta x}\right)$

and R are constants, which means that the velocity u varies with the square of r. Hence the velocity distribution across the section of a pipe is parabolic. This velocity distribution is shown in figure



To obtain an the ralationship between maximum velocity to average velocity

$$u = \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)(R^2 - r^2) \rightarrow (4)$$

The velocity is maximum when r=0.

Thus maximum velocity,

$$u_{max} = \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^2$$

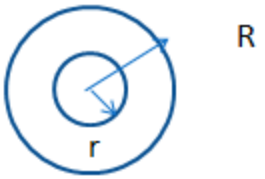
The average velocity v is obtained by dividing the discharge of the fluid across the section by the area of the pipe.

The discharge, Q across the section is obtained by considering the flow through a circular ring element of radius r and thickness dr.

The fluid flowing per second through this elementary ring is

dQ=velocity at a radius r x area of the ring element

$$dQ = \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)(R^2 - r^2) * (2\pi r dr)$$



integrating

$$Q = \int_0^R \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)(R^2 - r^2)\,(2\pi r dr)$$

$$Q = \frac{2\pi}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)\int_0^R (R^2 r - r^3)dr$$

$$Q = \frac{\pi}{2\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)\left(\frac{R^2 r^2}{2} - \frac{r^4}{4}\right)_0^R$$

$$Q = \frac{\pi}{2\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)\left(\frac{R^4}{2} - \frac{R^4}{4}\right)$$

$$Q = \frac{\pi}{2\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)\frac{R^4}{4} \qquad \Rightarrow Q = \frac{\pi}{8\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^4$$

Average velocity, V = Q/A

$$V = \frac{\frac{\pi}{8\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^4}{\pi R^2}$$

$$V = \frac{1}{8\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^2$$

$$\frac{u_{max}}{V} = \frac{\frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^2}{\frac{1}{8\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^2} = 2$$

The ratio of maximum velocity to average velocity is 2 if (i) the flow is laminar flow

(ii) the fluid is Newtonian fluid and

(iii) the flow takes place through a circular conduit.

Hagen – Poiseuille equation to find Pressure drop for a given length of pipe ($\Delta x=L$)

$$V = \frac{1}{8\mu}\left(\frac{P_1 - P_2}{L}\right)R^2$$

$8\mu VL=(P_1-P_2)R^2$

$P_1-P_2=8\mu VL/R^2$

$$P_1 - P_2 = \frac{8\mu VL}{\left(\frac{D}{2}\right)^2}$$

$$P_1 - P_2 = \frac{32\mu VL}{D^2}$$

- This is the pressure drop across a length of pipe L.

Head loss,

$$h_f = \frac{P_1 - P_2}{\rho g}$$

$$h_f = \frac{32\mu VL}{\rho g D^2}$$

- This equation is called Hagen – Poiseuille equation.

Relation between maximum velocity and average velocity for turbulent flow

The quantities N_{Re} and ratio V/u_{max} are useful in relating the average velocity to the maximum velocity in the centre of the tube as a function of flow conditions. An important method of measuring fluid flow is the pitot tube, which can be used to measure u_{max}, and this relationship is then used to determine the average velocity from this single observation.

For laminar flow V/u_{max} is exactly 0.5. The ratio changes rapidly from 0.5 to about 0.8 when laminar flow changes to turbulent flow.

Lubricating oil of relative density 0.86 is pumped through 305m of horizontal pipe having a diameter of 50 mm at the rate of $1.23 \times 10^{-3} m^3/s$. If the drop in pressure is 207 kN/m^2. Determine the absolute viscosity of oil assuming the flow to be laminar.

: Given $Q=1.23 \times 10^{-3} m^3/s$

$D= 50 mm = 0.05m.$

$A = \{D^2\}/4= (0.05)^2/4=1.963 \times 10^{-3} m^2.$

$V=Q/A = 1.23 \times 10^{-3}/1.963 \times 10^{-3} = 0.626m/$

s

$\Delta P=32\mu VL/D^2$

$$\mu = \frac{\Delta P D^2}{32VL} = \frac{207 * 10^3 * (0.05)^2}{32 * 0626 * 305} = 0.0847 \frac{N}{m^2 s}$$

A laminar flow is taking place in a pipe of diameter of 20 cm. The maximum velocity is 1.5m/s. Find the mean velocity and the radius at which this occurs. Also calculate the velocity at 4 cm from the wall of the pipe.

Sol : $V = \frac{1}{2} u_{max} = \frac{1}{2} * 1.5 = 0.75 m/s$

We have

$$u = \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)(R^2 - r^2) \rightarrow (1)$$

$$u_{max} = \frac{1}{4\mu}\left(\frac{P_1 - P_2}{\Delta x}\right)R^2 \rightarrow (2)$$

Dividing [1] by [2] $$\frac{u}{u_{max}} = \frac{R^2 - r^2}{R^2}$$

, substituting for u as v and solving for r, gives r = 0.0707m.

Again

$$\frac{u}{u_{max}} = \frac{R^2 - r^2}{R^2} \quad \Rightarrow u = u_{max}\left(\frac{R^2 - r^2}{R^2}\right)$$

$$u = u_{max}\left(\frac{R^2 - r^2}{R^2}\right)$$

$$1.5\left(\frac{0.1^2 - 0.06^2}{0.1^2}\right) = 0.96 m/s$$

A crude oil of viscosity 0.97 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 10 cm and of length 10m. Calculate the difference of pressure at the two ends of the pipe if 100kgs of the oil is collected in a tank in 30 seconds.

Sol: $\mu = 0.97 \text{poise} = 0.097\ Ns/m^2$

$\rho = 0.9*1000 = 900\ kg/m^3$

D= 10 cm = 0.1m

L= 10m

m=100/30 = 3.33kg/s

$Q = 3.33/900 = 3.7037*10^{-3}\ m^3/s$

$V = Q/A = 3.7037 *10^{-3}/(\pi* 0.1^2/4)$

V=0.4715m/s

$P_1 - P_2 = 32\mu VL/D^2 = 32*0.097*0.4715*10/0.1^2$

$P_1 - P_2 = 1463.75\ N/m^2$.

The velocity distribution in a pipe 25 mm diameter and 10m long in which the flow is laminar is given by $u_r = 2- cr^2$, where u_r is the velocity in m/s at any radius r and c is a constant. Determine (a) Rate of flow in lit/s (b) average velocity in m/s (c) maximum velocity in m/s (d) shear force at the pipe wall per m of pipe length in N (e) units of C and its value (f) the radial position at which a pitot tube be placed to measure the average velocity. Viscosity μ =0.001 $N.s/m^2$.

Sol : maximum velocity occurs when r =0

$u_r = 2 - c\, r^2$, when r = 0 ; $u_{max} = 2$ m/s (sol. for (c))

for laminar flow in a circular pipe $V/u_{max} = 0.5$

$V = 0.5\, u_{max} = 0.5 * 2 = 1$m/s (sol. for (b))

Rate of flow $Q = V * S = 1 * (\pi/4) * D^2 = 4.9087 * 10^{-4}$ m^3/s

Q =0.49087 lit/s (sol. for (a))

From Hagen – poiseuillies equation

$$P_1 - P_2 = \frac{32\mu VL}{D^2} = \frac{32 * 0.001 * 1 * 10}{0.025^2} = 512 \frac{N}{m^2}$$

$$\tau = \left(\frac{P_1 - P_2}{L} \right) \frac{r}{2} = \frac{512 * 0.025}{10 * 4} = 0.32N$$ Soln. For d

When r =R, $u_r = 0$

i.e., $0 = 2 - c R^2 \Rightarrow c = 2/R^2 = 12800\ m^{-1}s^{-1}$ (sol. for (e))

we know

or $$\frac{u}{u_{max}} = \frac{R^2 - r^2}{R^2}$$

$$\frac{V}{u_{max}} = 0.5 = \frac{R^2 - r^2}{R^2}$$

solving for r we get

$r = 8.83 * 10^{-3}m$

A liquid of specific gravity 4.7 and viscosity 1.3 centipoise flows through a smooth pipe of unknown diameter resulting in a pressure drop of 1270 N/m^2 for 2768 m length. What is the pipe dia if the mass flow rate is 75 kg/ sec?

Given Data

P1-P2= 1270 N/m^2

Length L= 2768 m , Density=4700kg/m^3

Viscosity=1.3×10^{-3} kg/m sec

$$\dot{m} = 75 kg/sec$$

$$P_1 - P_2 = \frac{32 \mu V L}{D^2}$$

$$\dot{m} = \rho V S$$

$$P_1 - P_2 = \frac{32 \mu V L}{D^2}$$

$1270=(32X\ 1.3x10^{-3}\ x\ V\ x2768)/D^2$

$D^2 = 0.0906V$

$$\dot{m} = \rho V S$$

$$\dot{m} = \rho V (\pi / 4) D^2$$

$$75 = 4700V\ (\pi/4)D^2$$

$1=15.66\ XVXD^2$

$1=15.66\ XVX0.0906V$

V=0.47m/
seD=0.2 m

What power is required per kilometer of a line to overcome the viscous resistance to the flow of glycerin through a horizontal pipe of diameter 10 cm at the rate of 10lit/s. kinematic viscosity 6 stokes,Relative density=1.333

$Q=10lit/s = 0.01m^3/s$

$D = 10cm = 0.1m \quad S = (\pi/4)*0.1^2 = 0.00785\ m^2$

$V = Q/S = 0.01/0.00785 = 1.27m/s$

From Hagen-Poiseullie's

$$\Delta P = \frac{32\mu vL}{D^2}$$

$$\Delta P = \frac{32 * 0.8 * 1.27 * 1000}{0.1^2}$$

$3251200N/m^2$

$$Power = \left(\frac{P_1 - P_2}{\rho} \right) \cdot m$$

$$= \left(\frac{\dot{m}}{\rho} \right) (P_1 - P_2)$$

Power,

$P = (P_1 - P_2) * Q = 3251200 * 0.01 = 32512W = 43.58HP$

HEAD LOSS IN fluid flow DUE TO FRICTION

Consider a horizontal pipe of radius R. The fluid flow is from left to right as shown in figure Let P_1 and P_2 be the intensity of the pressure at sections 1-1 and 2-2 respectively. Let sections 1-1 and 2-2 are separated by a length L.

Then the forces acting on the fluid element are

- The pressure force at section 1-1, $F_1 = P_1S$
- The pressure force at section 2-2, $F_2 = P_2S$
- The frictional resistance force, $F_3 = F\pi$ D L

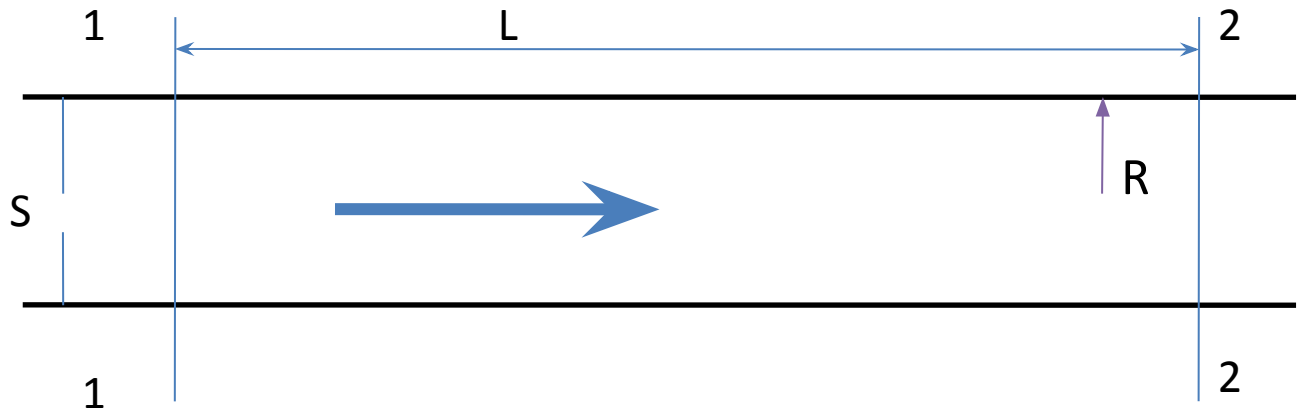
Where S is the cross sectional area of the pipe

D is the Diameter

As there is no acceleration, the summation of all forces in the direction of the flow must be equal to zero

$P_1 S – P_2 S – F \pi D L = 0$

$(P_1 – P_2) S = F \pi D L$



It was found by Darcy that the frictional resistance force per unit area is proportional to kinetic energy of fluid per unit volume

$$F \propto \frac{1}{2}\frac{mv^2}{V}, \text{where} V \text{ is volume}$$

$$F \propto \frac{1}{2}\rho v^2$$

$F = f \, ½ \, \rho v^2$, where f is the proportionality factor called friction factor

$(P_1 - P_2) \, S = f \, ½ \, \rho v^2 \pi \, D \, L$

$$P_1 - P_2 = \frac{f \rho v^2 \pi D L}{2S} = \frac{f \rho v^2 \pi D L}{2 \frac{\pi}{4} D^2}$$

$$P_1 - P_2 = \frac{4 f \rho v^2 L}{2D}$$

$$h_f = \frac{P_1 - P_2}{\rho g} = \frac{4 f v^2 L}{2 g D}$$

This equation is known as Darcy-Weisbatch equation.

Darcy friction factor is 4 times that of the Fanning friction factor

Relation between f and N_{Re}

We have from equation and

$$h_f = \frac{4 f v^2 L}{2 g D}$$

$$h_f = \frac{32\mu vL}{\rho g D^2}$$

from Hagen – Poiseuille equation.

Equating both the equations and solving for f

$$\frac{4 f v^2 L}{2gD} = \frac{32\mu vL}{\rho g D^2}$$

$$f = 16\frac{\mu}{\rho D v} = \frac{16}{N_{\mathrm{Re}}}$$

Hydraulically smooth pipe: In turbulent flow a rough pipe leads to a larger friction factor for a given Reynolds number than a smooth pipe does. If a rough pipe is smoothed, the friction factor is reduced. When further smoothing brings about no further reduction in friction factor for a given Reynolds number, the tube is said to be hydraulically smooth. Several idealized kinds of roughness is represented in figure.

The height of a single unit of roughness is denoted by k and is called roughness parameter. From dimensional analysis, f is found to be a function of both N_{re} and the relative roughness k/D, where D is the diameter of pipe. For a given kind of roughness, for eg. that shown in figure it can be expected that a different family of curve

Friction factor chart

For design purposes, the frictional characteristics of round pipe, both smooth and rough are summarized by friction factor chart which is a plot of log f vs log N_{re} . For laminar flow the f vs N_{re} relation is f =16/N_{re}. A log-log plot of the above equation is straight line with a slope of -1. This plot line is shown in figure for Reynolds number less than 2100.

N_{re} vs f would be found for each type of roughness. Experiments on artificially roughened pipe have confirmed these expectations. It has also been found that all clean, new commercial pipe seem to have the same type of roughness and that each material of construction has its own characteristic roughness parameter.

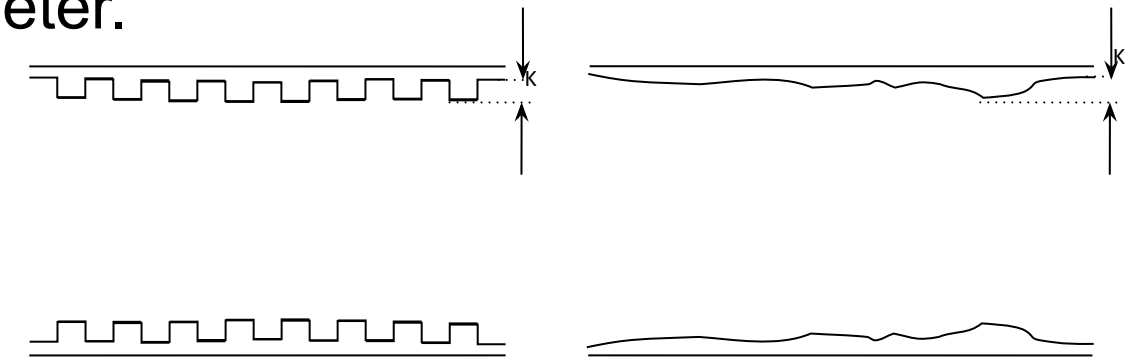


Figure 1 Idealized Hydraulically smooth pipes

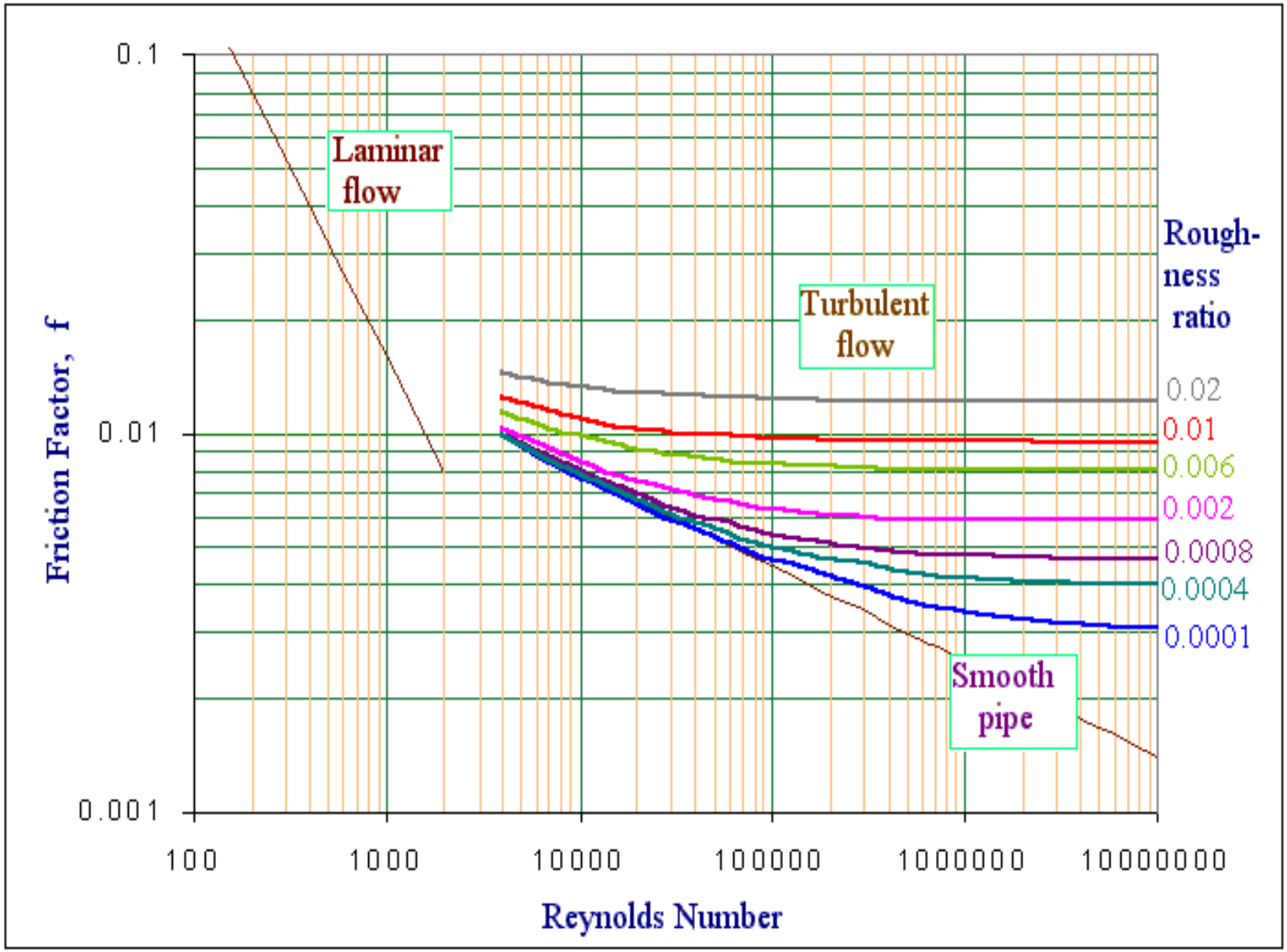
Old, foul and corroded pipe can be very rough, and the character of the roughness differs from that of clean pipe. Roughness has no appreciable effect on the friction factor for laminar flow unless k is so large that the measurement of the diameter becomes uncertain.

Material	Surface Roughness k, inch
Drawn tubing	0.00006
Commercial steel	0.0018
Galvanized iron	0.006
Cast iron	0.010
Wood stave	0.0072 - 0.036
Concrete	0.012 - 0.12
Riveted steel	0.036 - 0.36

Friction factor chart

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For turbulent flow the lowest line represents the friction factor for smooth tube. The other curved lines in the turbulent flow range represents the friction factors for various types of commercial pipe, each of which is characterized by a different value of k.



Fanning Friction Factor for Flow in Pipes.

For turbulent flow the lowest line represents the friction factor for smooth tube. The other curved lines in the turbulent flow range represents the friction factors for various types of commercial pipe, each of which is characterized by a different value of k.

The use of friction factor chart:

If the rate of flow, the density and viscosity of fluid and the diameter of the pipe are known, the Reynolds number is calculated. The Reynolds number is used as the abscissa, the friction factor is read as the ordinate and is used to calculate the desired friction loss or wall shear. The friction loss so calculated, combined with other forms of friction within the flow system, may then be substituted for h_f in the Bernoulli's equation.

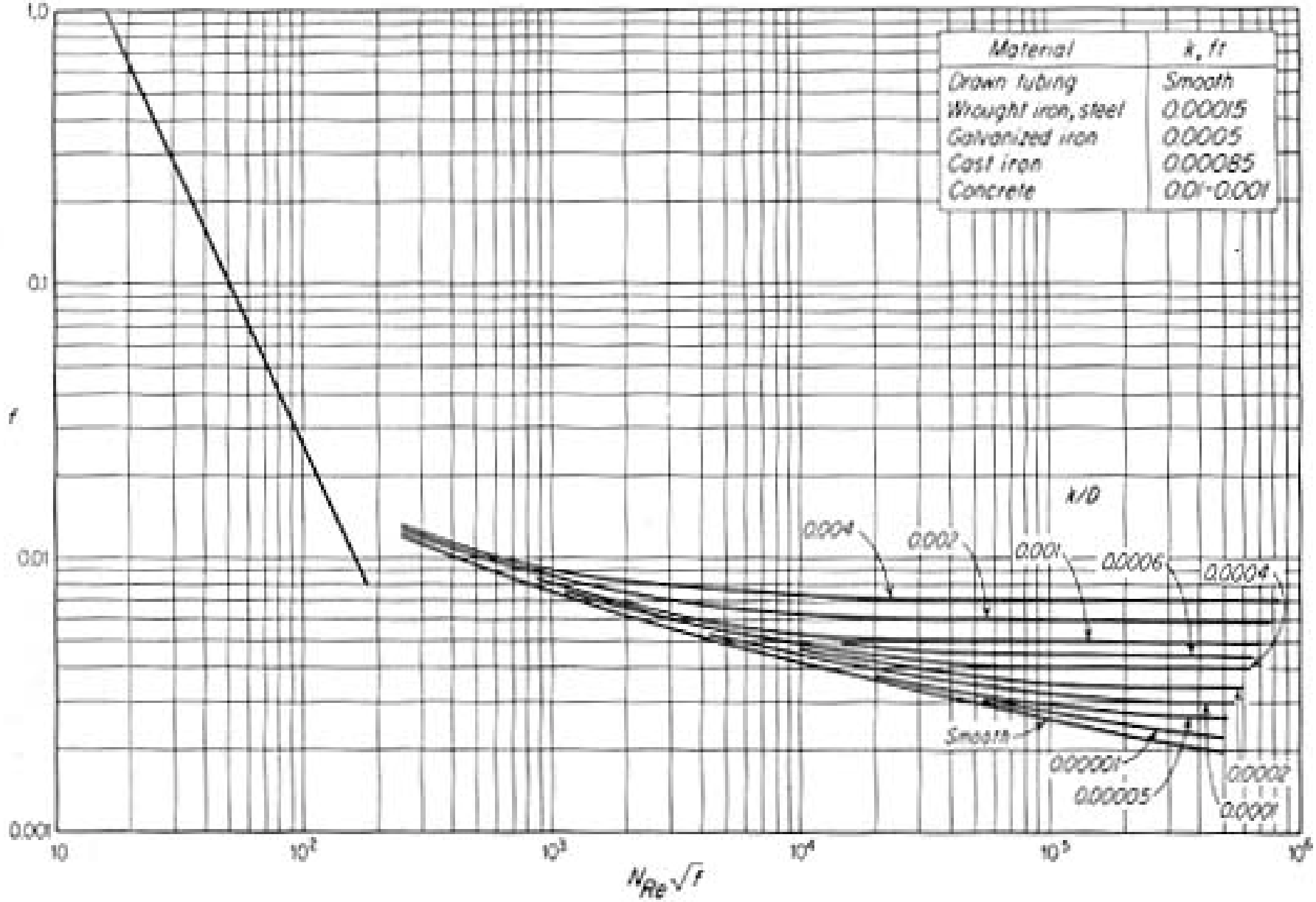


Figure 5-11 Friction-factor chart, $N_{Re}\sqrt{f}$ vs. f.

Calculation of flow rate from know skin friction

Friction factor chart is useful for calculating h_{fs} from a known pipe size and conditions of flow. It is not convenient for the calculation of the flowrate from a known h_{fs}, because ū, the average linear velocity, appears in both N_{re} and f. A trail and error solution can be used for this calculation, but a more convenient solution can be obtained by plotting $N_{re}\sqrt{f}$ as abscissa and f as ordinate.

$$N_{re}\sqrt{f} = \frac{D\bar{u}\rho}{\mu}\sqrt{\frac{\Delta P D}{2L\rho\bar{u}^2}} = \frac{D\rho}{\mu}\sqrt{\frac{\Delta P D}{2L\rho}}$$

To use figure, the abscissa is calculated from the known frictional loss (i.e., pressure loss), the ordinate f is read, and the velocity ū is calculated from f by means of equation

$$f = \frac{\Delta P D}{2L\rho\bar{u}^2}$$

Friction factor in flow through channels of non circular cross section

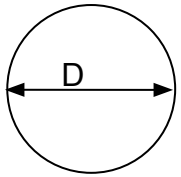
The friction in long straight channels of constant non circular cross section can be estimated by using the equations for circular pipe, if the diameter in Reynolds number and in the definition of the friction factor is taken as an *equivalent diameter*, defined as *four times the hydraulic radius*.

The hydraulic radius is denoted by r_H and in turn is defined as *the ratio of the cross sectional area of the channel to the wetted perimeter (perimeter of the channel in contact with fluid) of the channel.*

i. For circular pipes

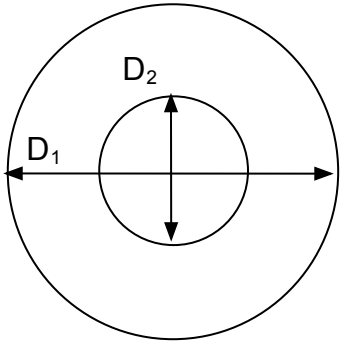
:

$$r_h = \frac{\text{cross sectional area}}{\text{wetted perimeter}} = \frac{\pi D^2/4}{\pi D} = \frac{D}{4}$$



The equivalent diameter = 4* r_H = 4*D/4 = D

ii. for Annulus

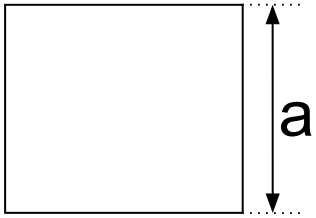


$$r_h = \frac{\text{cross sectional area}}{\text{wetted perimeter}} = \frac{(\pi D_1^2 - \pi D_2^2)/4}{\pi D_1 + \pi D_2} = \frac{D_1 - D_2}{4}$$

The equivalent diameter = $4 * r_H = 4*(D_1 – D_2)/4 = D_1 – D_2$

iii. For square with side 'a'

$$r_h = \frac{\text{cross sectional area}}{\text{wetted perimeter}} = \frac{a^2}{4a} = \frac{a}{4}$$



The equivalent diameter = $4 * r_H = 4*a/4 = a$

iv. For a rectangle with length l and breadth b

$$r_H = \frac{\text{cross sectional area}}{\text{wetted perimeter}} = \frac{l * b}{2(l + b)}$$



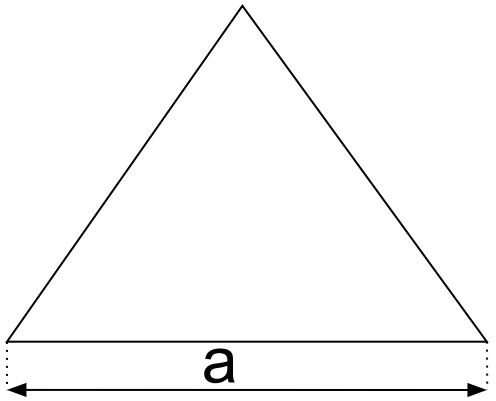
The equivalent diameter = 4* r_H

$$D_{eq} = \frac{2(l * b)}{(l + b)}$$

For a equilateral triangle

$$r_H = \frac{\frac{1}{2}a * a\sin 60}{3a} = \frac{a\sqrt{\frac{3}{2}}}{6} = \frac{\sqrt{3}}{12}a$$

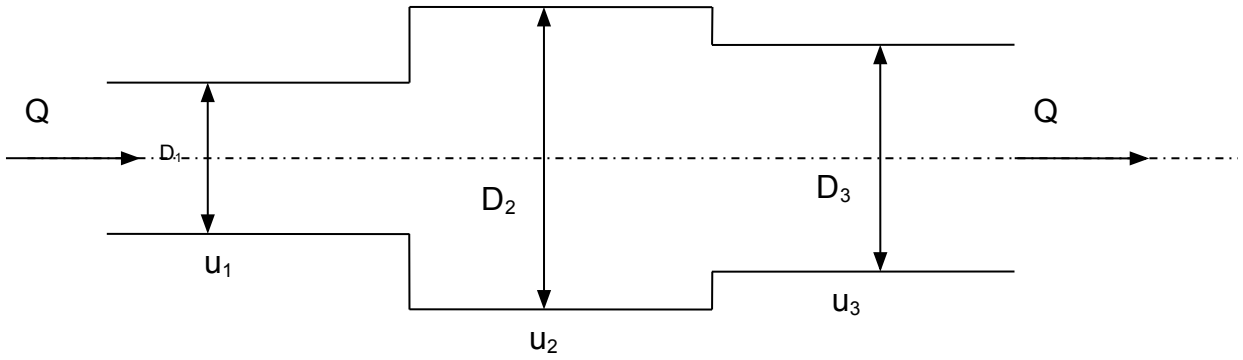
$$D_e = 4r_H = \frac{4\sqrt{3}}{12}a = \frac{a}{\sqrt{3}}$$



Flow through pipes
Pipes in series :

Q is the same in all the pipes (continuity equation)
i.e. $u_1 S_1 = u_2 S_2 = u_3 S_3 = Q$
so total $h_f = h_{f1} + h_{f2} + h_{f3}$

$$h_{f\,\text{total}} = \frac{4 f_1 L_1 v_1^2}{2gD_1} + \frac{4 f_2 L_2 v_2^2}{2gD_2} + \frac{4 f_3 L_3 v_3^2}{2gD_3}$$



A compound pipeline 1650m long is made of 45cm dia for 900m, 375cm dia for 450m and 30cm for 300m . It is required to be replaced by a pipe of uniform diameter. Find the diameter of the pipe.

For the equivalent pipe the head loss should be same

$$h_{f\,\text{total}} = \frac{4 f_1 L_1 v_1^2}{2gD_1} + \frac{4 f_2 L_2 v_2^2}{2gD_2} + \frac{4 f_3 L_3 v_3^2}{2gD_3}$$

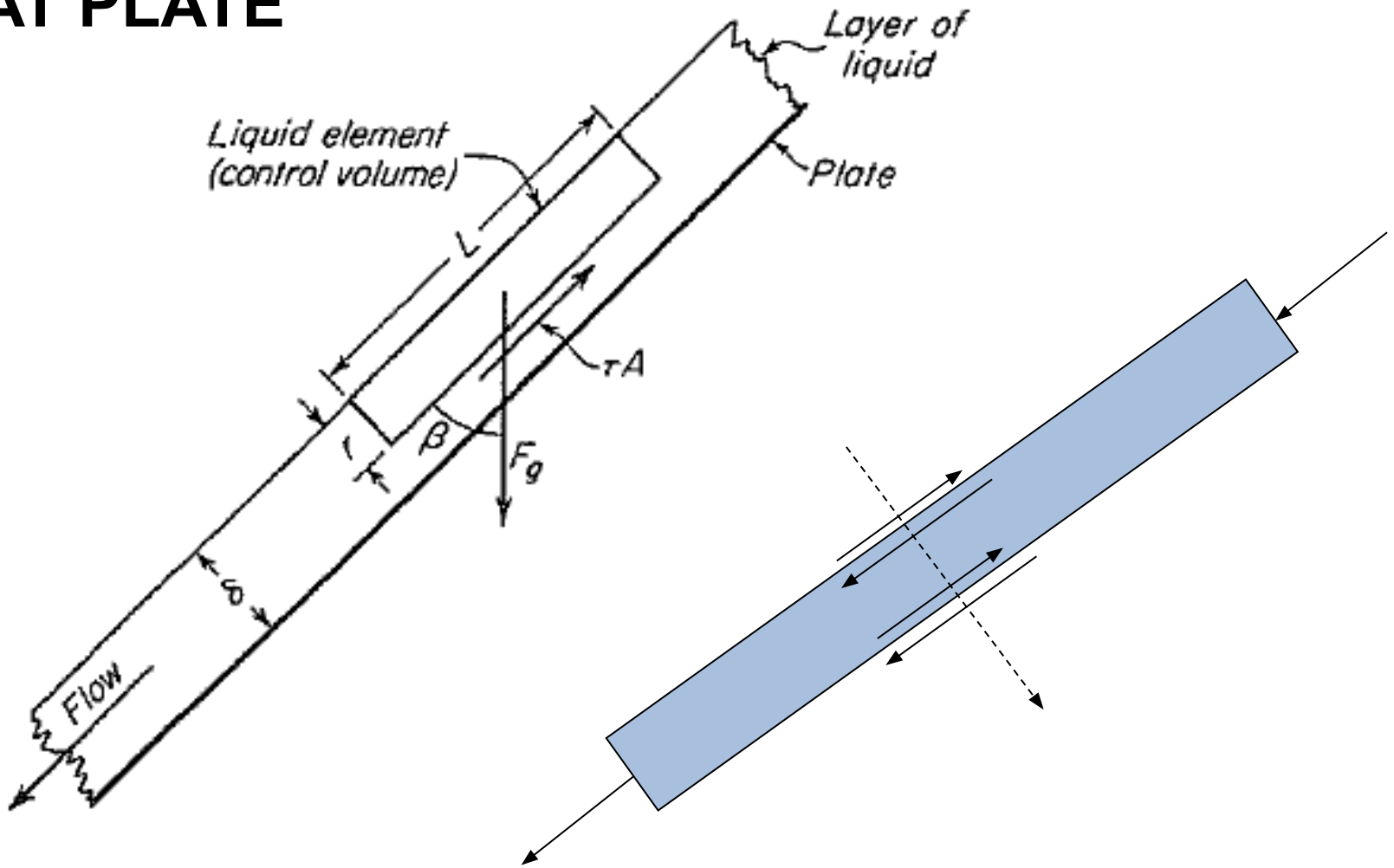
For same material $f_1=f_2=f_3$

$$h_{f\,\text{total}} = \frac{4 f}{2g}\left(\frac{L_1 v_1^2}{D_1} + \frac{L_2 v_2^2}{D_2} + \frac{L_3 v_3^2}{D_3}\right)$$

For equivalent pipe

$$h_{f\,\text{total}} = \frac{4f}{2g}(L_1 + L_2 + L_3)\frac{v^2}{D}$$

FLOW OF LIQUIDS IN THIN LAYERS OVER A FLAT PLATE



Forces on liquid element in Layer flow

FLOW OF LIQUIDS IN THIN LAYERS OVER A FLAT PLATE

- Consider a layer of a newtonian liquid flowing in steady flow at constant rate and thickness over a flat plate as shown in Fig
- The plate is inclined at an angle β with the vertical.
- The breadth of the layer in the direction perpendicular to the plane of the figure is b, and the thickness of the layer in the direction perpendicular to the plate is δ .
- Isolate a control volume as shown in Fig The upper surface of the control volume is in contact with the atmosphere, the two ends are planes perpendicular to the plate at a distance L apart, and the lower surface is the plane parallel with the wall at a distance r from the upper surface of the layer

- Since the layer is in steady flow with no acceleration, by the momentum principle the sum of all forces on the control volume is zero.
- The possible forces acting on the control volume in a direction parallel to the flow are the pressure forces on the ends, the shear forces on the upper and lower faces, and the component of the force of gravity in the direction of flow.
- Since the pressure on the outer surface is atmospheric, the pressures on the control volume at the ends of the volume are equal and oppositely directed. They therefore vanish.
- Also, by assumption, the shear on the upper surface of the element is neglected.

- The two forces remaining are therefore the shear force on the lower surface of the control volume and the component of gravity in the direction of flow. Then

$$F_g \cos\beta - \tau A = 0$$

F_g = gravity force
= shear stress on lower surface of control volume
A = area of lower surface of control volume
From this equation, noting that $A = bL$ and $F_g = \rho r L b g$

$$\rho r L b g \cos\beta = \tau L b \text{----------1}$$

$$\tau = \frac{\rho r g}{\cos\beta}$$

Since the flow is laminar

$$\tau = - \mu \frac{du}{dr}$$

$$- \mu \frac{du}{dr} = \rho r g \cos\beta$$

Rearranging and integrating between limits gives

$$(\delta^2 - r^2)----------2$$

δ = total thickness of the liquid layer

- Equation shows that in laminar flow on a plate the velocity distribution is parabolic, as it is in a pipe.
- The ratio of the maximum local velocity to the average velocity, however, is instead of 2.0 as in laminar flow in a pipe.

Consider now a differential element of cross-sectional area dS, where dS = b dr.

The differential mass flow rate *d*through this element equals *ρub dr.*

The total mass flow rate of the fluid then is

---------3

Substituting u from 2 into Eq. *3* and integrating gives

$(\delta^2 - r^2)bdr$

$= \Gamma$----------4

Γ

$=$

is called the liq̄uid loading dimension for Γ are kilograms per second per meter of width

Rearrangement of Eq.4 gives, for the thickness of the layer,

1/3

$$\delta = \sqrt[3]{\frac{3\mu\Gamma}{\rho^2 g \cos\beta}}$$

REYNOLDS NUMBER. The Reynolds number for layer flow is found by using four times the hydraulic radius as the length parameter. The hydraulic radius, defined as *S/Lp* , where S is the cross-sectional area of the layer and *Lp* is the wetted perimeter. For flow down a flat plate, S = *Lp δ,* and the hydraulic radius is equal to the layer thickness.

Reynolds number

$$N_{Re} = \frac{4r_H \bar{V} \rho}{\mu} = 4\delta \frac{\dot{m}}{\rho L_p \delta} \frac{\rho}{\mu} = \frac{4\Gamma}{\mu}$$

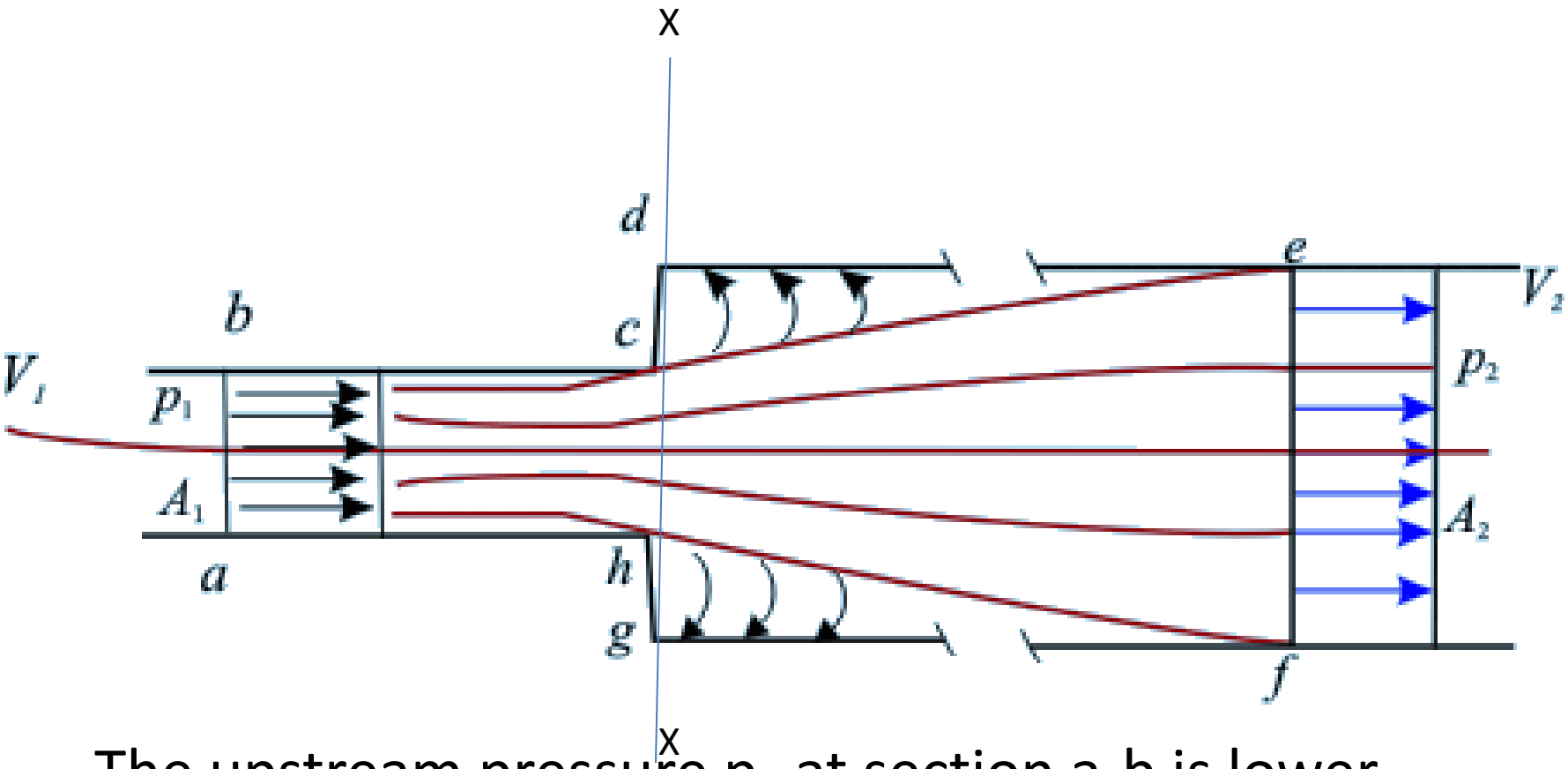
Friction loss from sudden expansion of cross section

If the cross section of the conduit is suddenly enlarged, the fluid stream separates from the wall and issues as a jet into the enlarged section. The jet then expands to fill the entire cross section of the larger conduit. The space between the expanding jet and the conduit wall is filled with fluid in vortex motion characteristic of boundary layer separation, and considerable friction is generated with in this space. This effect is shown in fig

Minor losses in flow systems

Minor losses include head loss or pressure drop due to pipe fittings, entrance and exit of pipe, sudden contraction or expansion etc. These are considered as minor because pressure drop is small compared to the major or frictional head loss. Minor Head losses are:

1. Head loss at the entrance of pipe
2. Head loss at the exit of pipe
3. Head loss due to obstruction in pipe
4. Head loss due to sudden expansion in pipe
5. Head loss due to sudden contraction in pipe
6. Head loss due to pipe fittings



The upstream pressure p_1 at section a-b is lower than the downstream pressure p_2 at section e-f

the upstream velocity

V_1 is higher than the downstream velocity

V_2 as a consequence of continuity

Consider a point at section X X just after the expansion.
The pressure remains as p_1
According to momentum equation ,
the external force acting = rate of change of momentum

$$(p_2 - p_1) A_2 = \rho Q (V_1 - V_2)$$

From the equation of continuity

$$Q = A_2 V_2$$

$$(p_2 - p_1) = \rho V_2 (V_1 - V_2)$$

Applying BE between 1 and 2

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

where h_L is the loss of head. Substituting for $(p_2 - p_1)$

$$h_L = \frac{\rho V_2(V_2 - V_1)}{\rho g} + \frac{V_1^2 - V_2^2}{2g}$$

$$h_L = \frac{2\rho V_2(V_2 - V_1) + \rho(V_1^2 - V_2^2)}{2\rho g}$$

$$h_L = \frac{2V_2V_1 + V_2^2 - V_1^2}{2g}$$

The friction loss h_{fe} from a sudden expansion of cross section is proportional to the velocity head of the fluid and can be written as

$$h_{fe} = k_e \frac{v_1^2}{2g}$$

, where k_e is a proportionality factor called expansion loss coefficient and is given by an emperical equation

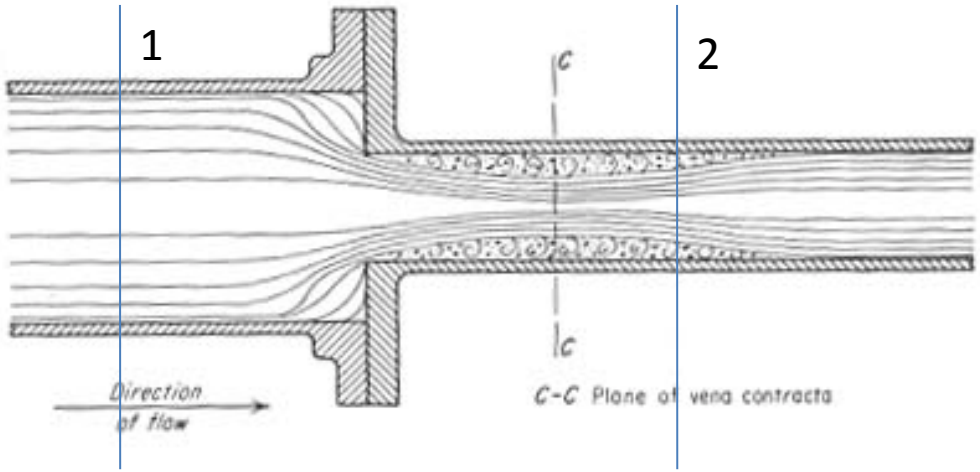
$$k_e = \left(1 - \frac{A_1}{A_2}\right)^2$$

and V_1 is the average velocity of the upstream.

Friction loss from sudden contraction of cross section

- When the cross section of the conduit is suddenly reduced, the fluid stream cannot flow around the sharp corner and the stream breaks contact with the wall of the conduit.
- A jet is formed, which flows into the stagnant fluid in the smaller section.
- The jet first contracts and then expands to fill the smaller cross section, and down stream from the point of contraction the normal velocity distribution eventually is established.
- The cross section of minimum area at which the jet changes from contraction to an expansion is called vena-contracta.
- The flow pattern of a sudden contraction is shown in figure

Flow at sudden contraction of cross section



C-C PLANE OF VENA CONTRACTA.

The friction loss from sudden contraction is proportional to the velocity head and can be calculated by the equation

$$h_{fc} = k_c \frac{v_2^2}{2g}$$

Where v_2 is average velocity in the downstream.

From section c to 2 the flow is like a sudden expansion, hence

=

$V_c = A_2V_2/Ac$ from equation of continuity

Cc = coefficient of contraction =Ac/A2

$$h_{fc} = \frac{(\frac{V_2}{C_c} - V_2)^2}{2g}$$

$$h_{fc} = \frac{(\frac{V_2}{C_c} - V_2)^2}{2g}$$

$$h_{fc} = [\frac{1}{Cc} - 1]^2 \frac{v_2^2}{2g}$$

$$h_{fc} = k_c \frac{v_2^2}{2g}$$

For laminar flow $k_c<0.1$ and the contraction loss h_{fc} is negligible.
For turbulent flow k_c is given by an empherical equation

$$k_c = 0.4\left(1 - \frac{A_2}{A_1}\right)$$

Head loss due to pipe fittings and valves

$$h_{ff} = k_f \frac{v_1^2}{2g}$$

where k_f is loss factor for fitting
V_1 average velocity in pipe leading to the fitting
Head loss at the entrance of a pipe

$$h_i = 0.5\frac{v^2}{2g}$$

Head loss at the exit of a pipe

$$h_o = \frac{v^2}{2g}$$

Fitting	Loss Coefficient, K
Gate valve (open to 75% shut)	0.25 - 25
Globe valve	10
Pump foot valve	1.5
Return bend	2.2
90º elbow	0.9
45º elbow	0.4
Large-radius 90º bend	0.6
Tee junction	1.8
Sharp pipe entry	0.5

Water flows at $7.2m^3/hr$ through a sudden enlargement from a 40mm to a 50mm diameter pipe. What is the loss in head.

Data: $Q = 7.2\ m^3/hr = 7.2/3600 = 0.002m^3/s$

D_a=40mm = 0.04m

D_b=50mm = 0.05m

$S_a = (\pi/4)* 0.04^2 = 1.26*10^{-3}m^2$

$S_b = (\pi/4)* 0.05^2 = 1.96*10^{-3}m^2$

We know $Q=u_aS_a \Rightarrow u_a=Q/S_a$

$u_a = 0.002/1.26*10^{-3} = 1.587m/s$

$$K_e = \left(1 - \frac{A_1}{A_2}\right)^2 = \left(1 - \frac{1.26*10^{-3}}{1.96*10^{-3}}\right)^2 = 0.12755$$

$$h_{fe} = K_e \frac{V_1^2}{2g} = 0.12755 \frac{1.587^2}{2*9.81}$$ =0.1637 m

h_{fe} = 0.01637*9.81 = 0.161 J/kg

A pipe of $0.1m^2$ area suddenly enlarges to $0.4m^2$ area. The quantity of water flowing in the pipe is $0.3m^3/s$ and the pressure at the smaller part of the pipe is $8.3*10^5$ N/m^2. Find out

i. head lost in m due to sudden expansion

ii. pressure in the larger part of the pipe

iii. power required in pumping the water through the enlargement

Data: $Q = 0.3m^3/s$

We know $Q=u_1A_1 \Rightarrow u_1=Q/S_a$

$u_1 = 0.3/0.1 = 3m/s$

and $u_b = 0.3/0.4 = 0.75m/s$

$A1 = 0.1m^2 \quad A2 = 0.4m^2$

$$K_e = \left(1 - \frac{A_1}{A_2}\right)^2 = \left(1 - \frac{0.1}{0.4}\right)^2 = 0.5625$$

$$h_{fe} = K_e \frac{V_1^2}{2g} = 0.5625 \frac{3^2}{2*9.81} = 0.258m$$

h_{fe} = 0.258 m

Applying Bernoulli's equation between inlet and outlet

$$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{u_2^2}{2g} + z_2$$

, but $z_1 = z_2$,

∴ the above equation reduces to $$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} = \frac{P_2}{\rho g} + \frac{u_2^2}{2g}$$

$$\frac{8.3*10^5}{1000}+\frac{3^2}{2}=\frac{P_2}{1000}+\frac{0.75^2}{2}$$

$P_2=8.3422*10^5$ N/m^2

Power, P= m * g *energy loss in head(metres)

$P = Q*\rho*g*h_{fe} = 0.3*1000*0.258*9.8=$**759.3W.**

PIPE, FITTINGS, AND VALVES

- ***Pipe and tubing*** Fluids are transported most often in pipe or tubing, which is circular in cross section and which is available in widely varying sizes, wall thickness, and materials of construction.
- There is no clear cut distinction between the terms pipe and tubing.
- Generally speaking, pipe is heavy-walled, relatively large in diameter, and comes in moderate lengths of 20 to 40 ft; tubing is thin walled and often comes in coils several hundred feet long.
- Metallic pipe can be threaded; tubing cannot. Pipe walls are usually slightly rough; tubing has very smooth walls. Lengths of pipe are joined by screwed, flanged, or welded fittings; pieces of tubing are connected by compression fittings, flare fittings, or soldered fittings.
- Finally, tubing is usually extruded or cold-drawn, while pipe is made by welding, casting, or piercing a billet in a piercing mill.

- Pipe and tubing are made from many materials, including metals and alloys, plastics, wood, ceramics, and glass. Most common is low carbon steel, fabricated into what is called “black iron pipe.” Wrought iron and cast-iron pipes are also used for a number of special purposes.
- Pipe and tubing are specified in terms of their diameter and their wall thickness.
- With steel pipe the standard nominal diameters range from 1/8 to 30 in.
- For large pipe, more than 12 in. in diameter, the nominal diameter are the actual outside diameters; for small pipe nominal diameter does not correspond to any actual dimension.

- The nominal value is close to the actual inside diameter for 3 to 12 in. pipe, but for very small pipe this is not true. Regardless of wall thickness, the outside diameter of all pipe of a given nominal size is the same, to ensure interchangeability of fittings.

- These standard sizes for steel pipe, therefore are known as IPS (iron pipe size) or NPS (normal pipe size).
- Thus the designation “2-in. nickel IPS pipe” means nickel pipe having the same outside diameter as standard 2-in. steel pipe.
- The wall thickness of pipe is indicated by the schedule number, given by the approximate equation

$$\text{Schedule number} = \frac{1000p'}{S}$$

where p’ = internal working pressure, lbf/in.^{2}

S = allowable stress, lbf/in.^{2}, for alloy used